LESSON PLAN · SCRATCH (GR 3–5) INTERMEDIATE: STORY, QUIZ & MUSIC STUDIO

Lesson 7: Operators: Maths in Code

Module 3 | 45 minutes | Operators and comparison | Grades 3–5 · ages 8–10

Objective: Use operator blocks (+ − × ÷ and join) to calculate, and check a typed answer against the right result.

Success criteria

- I can use operator blocks to calculate.
- I can check a typed answer with =.
- I can save an answer in a variable to use later.

Key vocabulary

- **operator**: A block that does maths or joins text.
- **expression**: A calculation, like (6) * (7).
- **compare**: Check if two values are equal, bigger or smaller.
- **join**: Put text and numbers together.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Human calculator (5 min): one learner is the “answer box” and can hold only one number card; the class discovers they need a second learner (a variable) to remember the first number.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	A times-table checker: learners follow the steps.
14 min	**Perform**	Code Lab: times-table checker, Inner blocks first
9 min	**Probe**	Code Lab: adding machine, activity
4 min	**Wrap-up**	Exit ticket: Write the condition that checks if the answer equals 8 × 9.

Unplugged warm-up

Human calculator (5 min): one learner is the “answer box” and can hold only one number card; the class discovers they need a second learner (a variable) to remember the first number.

Teaching the idea

Operator blocks are reporters that compute values and nest inside other blocks; inner blocks are evaluated first. Typed answers can be compared with = and used in arithmetic. Because the answer reporter is overwritten by each ask, earlier answers must be saved in variables.

Analogy: Operator blocks are like a calculator built into your sentence: the sprite works out the sum before speaking.

Live demo: Build the calculator and show the bug of using “answer + answer” (it doubles the second number). Fix it with a variable.

Questions to ask

Ask	Listen for
Which block is worked out first?	The innermost one.
Why did the calculator give the wrong total?	The first answer wasn’t saved.
How do you write × in Scratch?	With the * operator.

Differentiation

Support
Provide the if-else with the condition already built; learners change the question and numbers.

Stretch
A calculator that asks which operation to do (+ or −) and uses if-else to choose.

Common misconceptions

Misconception	How to fix it
answer remembers every reply	It only holds the latest; save earlier ones in variables.
Operators go in the script like normal blocks	They go inside slots of other blocks.
Order doesn’t matter in nested sums	Inner blocks are calculated first.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Write the condition that checks if the answer equals 8 × 9.

Homework
Write three maths questions you could put in a quiz, with their answers.

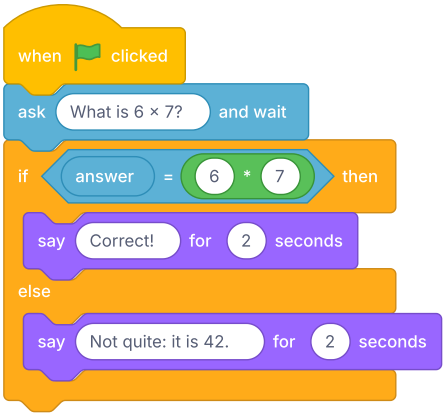
Answer key — Kitabash activities

1. Code Lab: times-table checker (perform · code lab)

Goals checked: 42 is marked correct · A wrong answer gets help

One solution:

Host



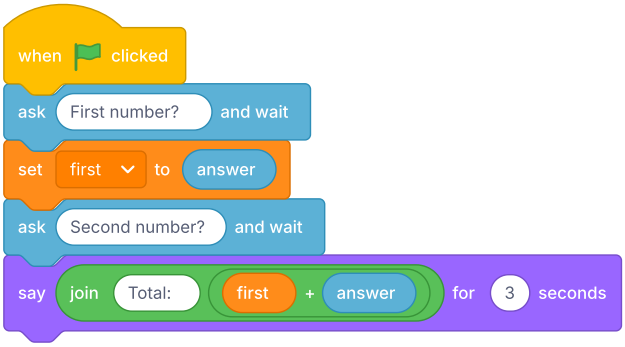
2. Inner blocks first (perform · code output)

3. Code Lab: adding machine (probe · code lab)

Goals checked: Adds 3 and 4 to make 7 · Adds 10 and 25 to make 35 · Uses join for a sentence

One solution:

Host



4. Explain your thinking (probe · free response)

Look for: variable, answer, replace. **Example:** The answer reporter only remembers the last thing typed. When we ask the second question, the first answer is replaced, so we save it in a variable first.

Answer key — worksheet

1. **Key words:** operator = A block that does maths or joins text. · expression = A calculation, like (6) * (7). · compare = Check if two values are equal, bigger or smaller. · join = Put text and numbers together.
2. **Plan your code:** Goals: 42 is marked correct; A wrong answer gets help.
3. **Inner blocks first:** A. 4
4. **Plan your code:** Goals: Adds 3 and 4 to make 7; Adds 10 and 25 to make 35; Uses join for a sentence.
5. **Explain your thinking:** The answer reporter only remembers the last thing typed. When we ask the second question, the first answer is replaced, so we save it in a variable first.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.